The Time Course of Irradiation Changes in Proton Beam-treated Uveal Melanomas

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Background: Although histopathologic changes in uveal melanomas treated by irradiation have been well characterized, the timing of these effects after irradiation has not been previously evaluated.

Methods: The series consisted of a total of 92 eyes with uveal melanoma, enucleated at various intervals after proton irradiation (range, 1 month to 8 years) due to complications (n = 66) or tumor growth (n = 22), or at autopsy (n = 4). Study slides were read, using a standard protocol, by two pathologists masked to the timing of the enucleation after irradiation and the reason for the surgery.

Results: The prevalence of inflammation decreased whereas fibrosis increased with time from 15% of cases enucleated within 12 months (early cases) to 61% of those enucleated more than 30 months (late cases) after irradiation (*P* for trend = 0.0005). Tumor necrosis and blood vessel damage occurred early, and the prevalence of these changes was constant over time. Excluding tumors with evidence of growth after irradiation, mitotic figures became progressively less common as the interval between irradiation and enucleation increased (*P* for trend = 0.015); no mitotic figures were present in 40 high-power fields more than 30 months after irradiation.

Conclusion: Histopathologic changes in irradiated melanomas vary according to the time elapsed since irradiation. Inflammation decreases with time whereas fibrosis becomes more prevalent with time after irradiation. Among controlled tumors, mitotic figures appear only in recently irradiated tumors. *Ophthalmology 1993;100:1555–1560*

In a previous investigation,[1] we identified several histopathologic features that distinguish irradiated from nonirradiated uveal melanomas. Compared with a matched series treated by enucleation, irradiated tumors were more likely to have necrotic and fibrotic changes, inflammatory infiltrates, and greater numbers of balloon cells. Additionally, in the irradiated series, there were fewer mitotic figures counted in 40 high-power fields and a higher prevalence of tumor blood vessel damage. These findings suggest an effect of irradiation on melanoma tissue that includes damage to both tumor cells and blood vessels, and an altered capacity for cellular reproduction.

In this investigation, we studied the timing of these changes by evaluating a group of eyes enucleated because of complications or tumor regrowth at variable intervals after proton irradiation. To our knowledge, this is the first study in which the timing of irradiation effects in a large series of uveal melanomas has been evaluated systematically.

Subjects and Methods

Proton beam irradiation has been used in the treatment of uveal melanoma at the Harvard Cyclotron since July

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